

Sorting & Classifying

Lesson Plan — AI & Digital, Grade 2

Subject:	AI & Digital Literacy
Grade Level:	Grade 2
Topic:	Sorting and Classifying
Duration:	40–45 minutes
Materials Needed:	Crayons, toy blocks/animals/plants, fruit pictures or plastic fruit, books, pens, paper, whiteboard or chart paper

Learning Objectives

By the end of this lesson, students will be able to:

- Explain what AI and digital technology mean in simple terms.
- Define sorting as putting things in order by one shared feature (colour, size, shape, or number).
- Define classifying as grouping things by what they are (what makes them similar).
- Give real-life examples of sorting and classifying at home and at school.
- Identify how computers and AI use sorting and classifying every day (e.g., sorting emails, classifying photos).

Lesson Flow

1. Warm-Up: What is AI & Digital? (5 minutes)

Begin by asking students: "Have you ever used a tablet or phone to play a game or watch a video?" Explain that this is called **digital** technology. Then introduce **AI**: AI means teaching computers to think and learn — computers can look at things, sort them, and make smart choices, just like we do!

2. Meet Sorting (8 minutes)

Introduce the idea: **Sorting means putting things in order**. Use the crayon example — sorting crayons by colour (red, orange, yellow, green, blue, violet). Explain that when we sort, we look at one thing that is the same, such as colour, size, or shape.

Discuss real-life examples:

- Sorting toys by size — from smallest to biggest!
- Sorting books by height on the shelf.
- Sorting fruits by type — apples, bananas, and mangoes in separate groups.

3. Meet Classifying (8 minutes)

Introduce the idea: **Classifying means grouping things by what they are**. Give the example of putting all animals in one group and all plants in another. When we classify, we look for what makes things similar.

Discuss real-life examples:

- Grouping clothes by season — summer clothes together, winter clothes together!
- Grouping school supplies by use — pens, books, and colours each in their own group.

4. Sorting vs. Classifying — Spot the Difference (5 minutes)

Sorting	Classifying
Putting things in order — by colour, size, or number.	Grouping things by what they are — like animals, plants, or toys.

Both sorting and classifying help us — and computers — stay organised!

5. How Does AI Use Sorting & Classifying? (5 minutes)

Explain that computers use sorting and classifying every single day:

- AI sorts emails so important ones come first!
- AI classifies photos into animals, people, or places.

Quiz Time: Sorting Game (5 minutes)

Ask students: **Which of these is sorted correctly?**

Red, orange, yellow, green, blue (colours in rainbow order)	✓ Correct
5, 2, 8, 1, 4 (numbers jumbled, not in order)	✗ Incorrect
Fruits arranged from smallest to biggest	✓ Correct

Discuss why each example is sorted correctly or incorrectly, and have students explain their reasoning out loud.

Hands-On Activities

Activity 1: Sorting Activity (in class or at home)

Try sorting toys or clothes! Follow these simple steps:

- 1 Pick a category to sort by (colour, size, or type).
- 2 Put things in order or groups based on that category.

Activity 2: Classifying Activity (in class or at home)

Group books or snacks by kind! Follow these simple steps:

- 1 Look at your things carefully.
- 2 Put similar things together in the same group.

Why Do We Sort & Classify?

- Helps us find things quickly!
- Helps us keep things neat and tidy!

Sorting and classifying are super useful skills! When we sort and classify, we help ourselves AND we help computers too. AI uses sorting and classifying to understand information and make smart decisions every day.

Wrap-Up Summary (4 minutes)

- Sorting puts things in order — like crayons by colour or numbers in a row.
- Classifying puts things in groups — like animals together and plants together.
- Both help us and computers stay organised and find things quickly.
- AI uses sorting and classifying every single day!

Assessment Ideas

- Observe students during the sorting and classifying activities — can they explain their grouping rule?
- Use the quiz questions as an exit check (thumbs up/down for correct vs. incorrect sorting examples).
- Ask each student to name one example of sorting and one example of classifying from their own life.